

Lab sheet 17**Submission Due: 21.10.2024****Title:** Introduction to React and Basic Functional Components**Aims:**

- Setting up development environment
- Project creation
- Understanding the core concepts

Tasks:

- Install Node.js and npm
- Setting up React
- Start the development server
- Basic React Components

Activities :**1. Install Node.js and npm****Windows:**

- Download the appropriate Node.js installer from the official website:
<https://node.js.org/en/download/>
- Run the downloaded installer file and follow the on-screen instructions.
- Verify the installation by opening a command prompt and typing
node -v and npm -v

macOS:

- Install Homebrew if you don't already have it: <https://docs.brew.sh/Installation>
- Once Homebrew is installed, run the following command to install Node.js:
brew install node
- Verify the installation by opening a Terminal window and typing
node -v and npm -v

Linux:

- The installation method for Linux depends on your specific distribution. However, generally, you can use your package manager to install Node.js. For example, on Ubuntu, you can use the following command:
sudo apt install nodejs npm
- Verify the installation by opening a terminal window and typing
node -v and npm -v

2. Setting up React

```
E:\SEUSL\AS-02\SWT22022\NEW\React>npm create vite@latest
✓ Project name: ... demo
✓ Select a framework: » React
✓ Select a variant: » JavaScript

Scaffolding project in E:\SEUSL\AS-02\SWT22022\NEW\React\demo...

Done. Now run:

  cd demo
  npm install
  npm run dev

E:\SEUSL\AS-02\SWT22022\NEW\React>
```

3. Start the development server

Install the required dependencies for the project:

npm install

Run the development server:

```
E:\SEUSL\AS-02\SWT22022\NEW\React\demo>npm run dev

> demo@0.0.0 dev
> vite

VITE v5.4.8 ready in 1291 ms

  Local:   http://localhost:5173/
  Network: use --host to expose
  press h + enter to show help
```

4. Basic React Components

Hello.jsx

```
function Hello() {

  return (
    <>
      <div>
        <h2>Hello World !</h2>
      </div>
    </>
  )
}
```

```
}  
  
export default Hello
```

main.jsx

```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
// import App from './App.jsx'  
import './index.css'  
import Hello from './hello.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <Hello />  
  </StrictMode>,  
)
```

Discussion:

- React Components